Technical Report

CMPG 325 Computer Networks | Individual Semester Project

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Project ID	CMPG325-2026-107
Client	CLI-107: Kalahari Biodiversity Research Centre (Upington)
Industry	Research
Assigned addressing block	192.168.48.0/24
Assigned networking challenge	DNS (internal name resolution service)
Design constraint	No network downtime tolerated during month-end processing
Change request	CR11: Management adopts VoIP handsets; voice traffic must be accommodated on the existing design

1. Introduction

This report documents the design, implementation and testing of a network for the Kalahari Biodiversity Research Centre in Upington, built and simulated in Cisco Packet Tracer. The centre combines administrative work, including month-end processing, with research and laboratory work. The network must connect these groups, provide internal name resolution (DNS), and accommodate VoIP handsets introduced by change request CR11, within the assigned address block 192.168.48.0/24.

2. Client Requirements

- Separate connectivity for Administration, Research, Servers and Voice, so that each group has appropriate access without sharing unrelated traffic.
- An internal DNS service so staff can reach internal resources by name instead of by IP address (assigned challenge).
- Addressing built from the assigned block 192.168.48.0/24.
- Support for VoIP handsets without degrading existing data services (CR11).
- No network downtime during month-end processing (design constraint).
- A working, testable Packet Tracer implementation with documented evidence.

3. Network Design

The network uses a router-on-a-stick design. One router (Cisco 2911) connects to a core switch (Switch0). Two access switches (Admin Switch and Research Switch) connect to the core switch. The core switch also connects the server, which runs both DNS and DHCP. Each access switch serves three PCs and one IP phone.

Device	Model	Role	Quantity
Router0	Cisco 2911	Inter-VLAN routing (router-on-a-stick)	1

Device	Model	Role	Quantity
Switch0	Cisco 2960-24TT	Core switch; trunks to router and access switches	1
Admin Switch, Research Switch	Cisco 2960-24TT	Access switches for each department	2
DNS-DHCP-SERVER	Server-PT	DNS and DHCP services (VLAN 30)	1
AdminPC1-3, ResearchPC1-3	PC-PT	Staff computers	6
Admin phone, Research phone	Cisco 7960 IP phone	VoIP handsets (CR11)	2

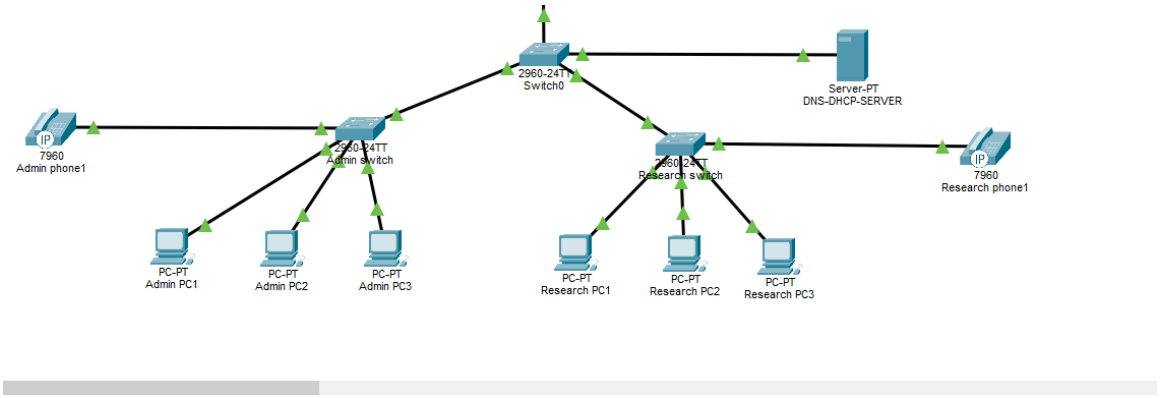


Figure 1: Packet Tracer topology showing all devices and cabling.

3.1 Design decisions

- **Four VLANs** (Admin, Research, Servers, Voice) separate departments, so a fault in one group stays within that group.
- **Router-on-a-stick** was chosen because it needs only one router link and is simple to explain and test.
- **Server on its own VLAN** keeps DNS and DHCP separate from user traffic.
- **Voice on its own VLAN** lets the phones be added without adding load to the Admin data network (CR11).

4. IP Addressing Plan

The assigned /24 block was split into four equal /26 subnets, one per VLAN. Equal subnets were chosen for simplicity and ease of explanation, because the assigned challenge is DNS rather than address efficiency. Each /26 provides 62 usable addresses, which is more than enough for the current devices and leaves room for growth.

VLAN	Department	Subnet	Gateway (router)	Usable range
10	Admin	192.168.48.0/26	192.168.48.1	192.168.48.1 - .62
20	Research	192.168.48.64/26	192.168.48.65	192.168.48.65 - .126
30	Servers	192.168.48.128/26	192.168.48.129	192.168.48.129 - .190
40	Voice	192.168.48.192/26	192.168.48.193	192.168.48.193 - .254

The server uses a static address of 192.168.48.130 (mask 255.255.255.192, gateway 192.168.48.129). All PCs and phones receive their addresses by DHCP.

5. Implementation

5.1 VLANs and switch ports

VLANs were created on each switch. On the access switches, ports Fa0/1 to Fa0/3 are access ports in the department VLAN (VLAN 10 on Admin Switch, VLAN 20 on Research Switch), and port Fa0/4 carries the phone using the voice VLAN (VLAN 40). On the core switch, the links to the router and the two access switches are 802.1Q trunks, and the server port is an access port in VLAN 30.

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name admin
Switch(config-vlan)#exit
Switch(config)#vlan 40
Switch(config-vlan)#name voice
Switch(config-vlan)#exit
Switch(config)#interface range fastEthernet 0/1 -3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#interface fastethernet 0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport voice vlan 40
Switch(config-if)#exit
Switch(config)#
```

Figure 2: Admin Switch VLAN and port configuration.

```
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up


Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#name research
Switch(config-vlan)#exit
Switch(config)#vlan 40
Switch(config-vlan)#name voice
Switch(config-vlan)#exit
Switch(config)#interface range fastEthernet 0/1 - 3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#interface fastEthernet 0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport voice vlan 40
Switch(config-if)#exit
Switch(config)#
```

Figure 3: Research Switch VLAN and port configuration.

5.2 Inter-VLAN routing

The router's single physical link to the core switch is divided into four sub-interfaces (Gi0/0.10, .20, .30 and .40). Each uses 802.1Q encapsulation for its VLAN and holds the gateway address for that subnet.

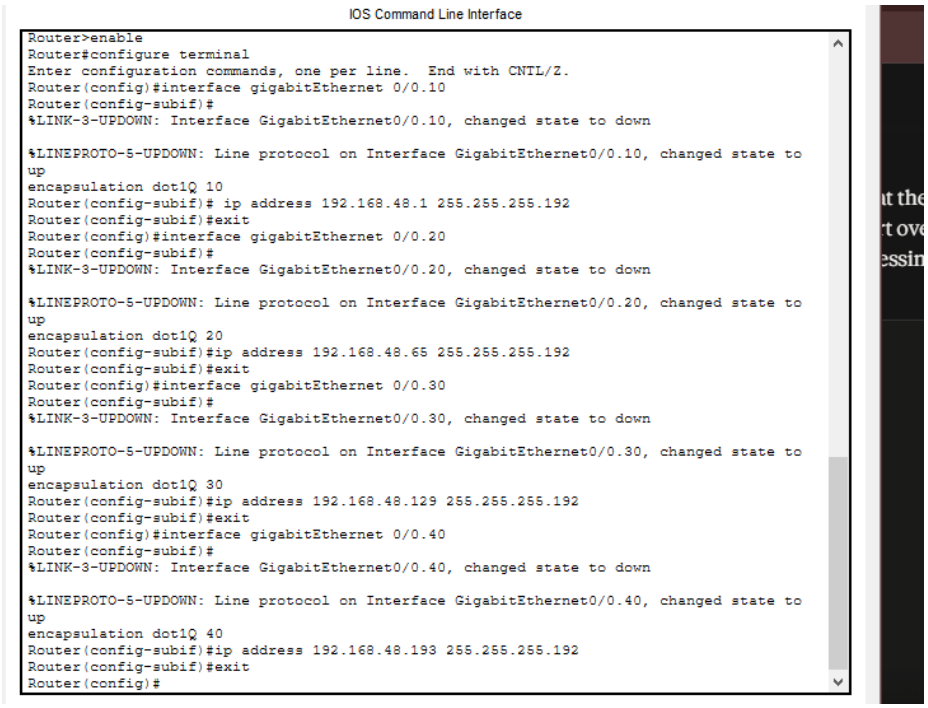


Figure 4: Router sub-interface configuration for VLANs 10, 20, 30 and 40.

5.3 DHCP

The server provides one DHCP pool per VLAN, each with the correct gateway and the server as DNS server. Because DHCP requests are broadcasts that do not cross VLANs, the router forwards them to the server using **ip helper-address 192.168.48.130** on the Admin, Research and Voice sub-interfaces. The Servers VLAN does not need relay, since the DHCP server is on that VLAN.

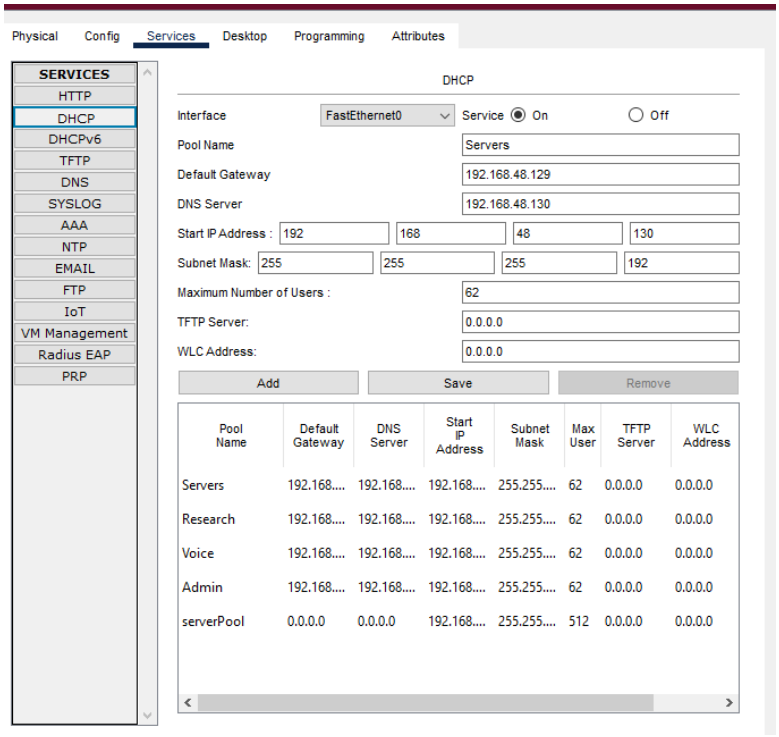


Figure 5: DHCP pools configured on the server.

6. Assigned Networking Challenge: DNS

What was configured. The DNS service was enabled on the server (192.168.48.130) and an A record was added for **server.kbrc.local** pointing to 192.168.48.130. The DHCP pools hand out the server's address as the DNS server, so every client is configured to use it automatically.

Why it is appropriate. Staff can reach internal resources by name rather than by remembered IP addresses, and if an address changes only the DNS record needs updating. Placing DNS on the same server as DHCP, in a protected Servers VLAN, keeps the services central and easy to manage.

How it was verified. Pinging server.kbrc.local from a PC in the Admin VLAN and from a PC in the Research VLAN resolved the name to 192.168.48.130 and received replies with 0% loss.

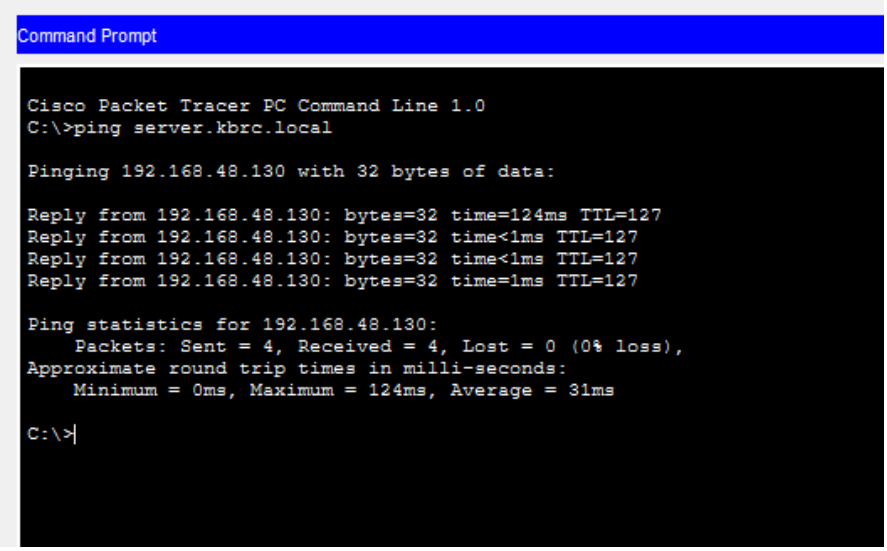


Figure 6: DNS test from AdminPC1: name resolves to 192.168.48.130.

7. Change Request CR11: VoIP Handsets

The existing design accommodates the phones without redesign. Each access switch has one port set with the voice VLAN (VLAN 40), and a Voice DHCP pool serves the phones. The 2960 switches do not supply Power over Ethernet, so a power adapter module was added to each phone. Both phones received addresses from the Voice pool (192.168.48.194 and 192.168.48.195), appear in the router's ARP table on the Voice sub-interface, and replied to pings from the router with 100% success.

```
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!


%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.40, changed state to up


Router>enable
Router#show ip arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.48.2            13  0001.96AD.CC07  ARPA   GigabitEthernet0/0.10
Internet  192.168.48.3            61  0001.6388.C852  ARPA   GigabitEthernet0/0.10
Internet  192.168.48.67           62  0002.4A25.61B4  ARPA   GigabitEthernet0/0.20
Internet  192.168.48.130          74  0010.11C5.7120  ARPA   GigabitEthernet0/0.30
Internet  192.168.48.194          13  00D0.9797.43C3  ARPA   GigabitEthernet0/0.40
Internet  192.168.48.195          12  0001.97AD.1171  ARPA   GigabitEthernet0/0.40
Router#
```

Figure 7: Router ARP table showing both phones on the Voice VLAN (Gi0/0.40).

```
Internet  192.168.48.195          12   0001.97AD.1171  ARPA   GigabitEthernet0/0.40
Router#ping 192.168.48.194

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.48.194, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/6/16 ms

Router#ping 192.168.48.194

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.48.194, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#
```

Figure 8: Router pings to the phone at 192.168.48.194: 5/5 replies.

8. Month-End Downtime Constraint

The client does not tolerate downtime during month-end processing. The design reduces the risk in these ways:

- **VLAN separation:** a problem in one department, such as a faulty PC or phone, stays inside its VLAN and does not affect Admin traffic.
- **Dedicated Servers VLAN:** DNS and DHCP are kept separate, so staff can still resolve names and obtain addresses.
- **Voice on its own VLAN:** adding phones does not add load to the Admin data network.
- **Tested configuration:** every VLAN, trunk, DHCP pool and DNS record was tested before sign-off.

Limitations. Router0 and Switch0 are single devices, so if either failed, the network would go down. For a production deployment, the recommended improvements are a second router with a redundancy protocol such as HSRP, a second core switch, and redundant links between switches using spanning tree. These were not built, to keep the project within the scope of the brief.

9. Testing and Evidence

Test	Expected	Result
DHCP on Admin PCs (VLAN 10)	Address in 192.168.48.2-.62, gateway .1	Passed (192.168.48.2 to .4)
DHCP on Research PCs (VLAN 20)	Address in 192.168.48.66-.126, gateway .65	Passed (192.168.48.66 to .68)
DNS from Admin VLAN	server.kbrc.local resolves to 192.168.48.130	Passed (4/4 replies)
DNS from Research VLAN	server.kbrc.local resolves to 192.168.48.130	Passed (4/4 replies)
Cross-VLAN routing (Research to Admin)	Ping to 192.168.48.3 succeeds	Passed (first packet lost while devices learned each other, then replies)
Phones on Voice VLAN	Addresses in 192.168.48.192/26	Passed (.194 and .195, router ping 5/5)

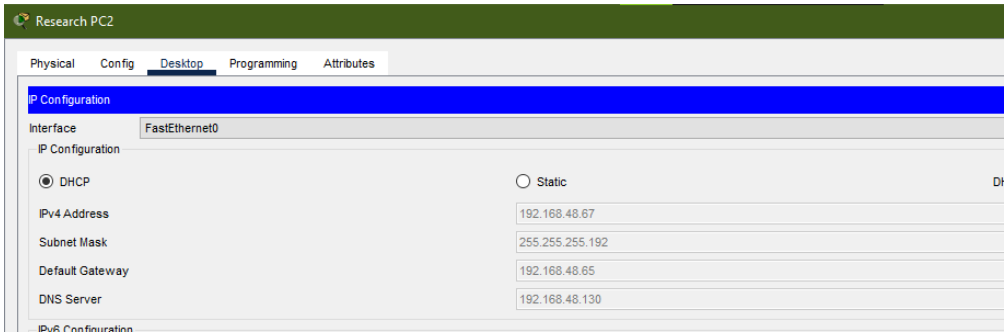


Figure 9: ResearchPC2 received 192.168.48.67 by DHCP (gateway .65, DNS .130).

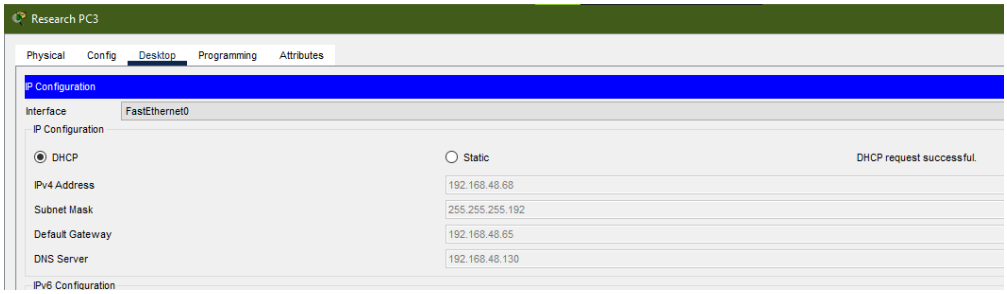


Figure 10: ResearchPC3 received 192.168.48.68 by DHCP.

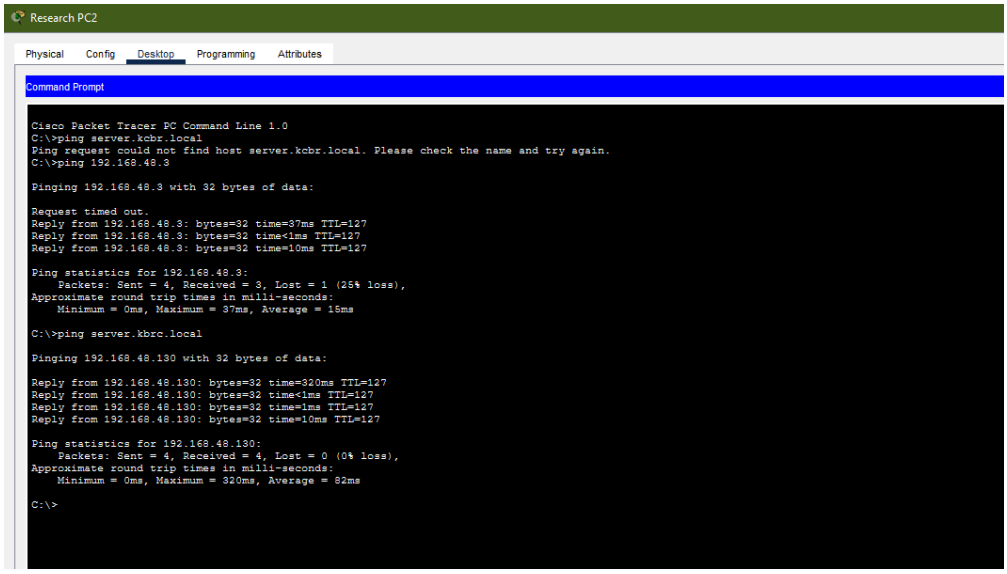


Figure 11: ResearchPC2: DNS resolves from the Research VLAN, and Research reaches Admin through the router.

10. Troubleshooting

Problem 1: DHCP failed (APIPA address). AdminPC1 showed the message "DHCP failed. APIPA is being used" and an address starting 169.254.

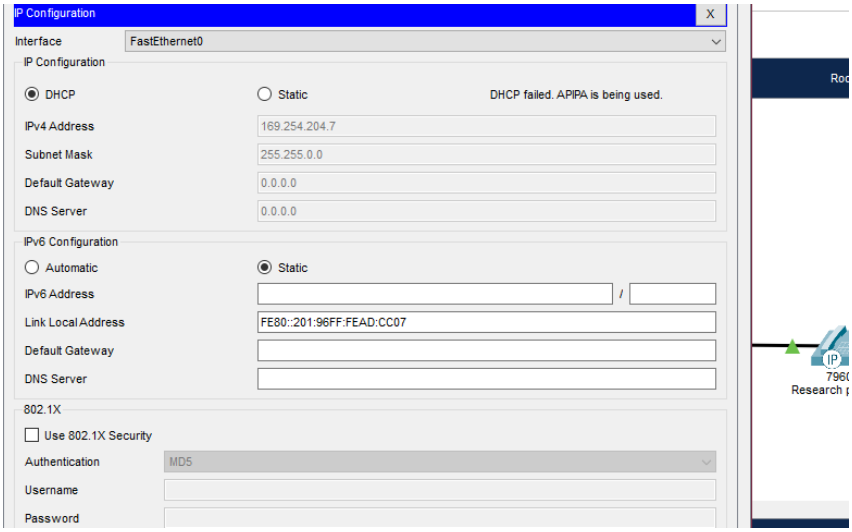


Figure 12: AdminPC1 failed to get an address by DHCP.

Diagnosis. The command **show interfaces trunk** on Switch0 showed that only VLANs 1 and 30 were active on the trunks. VLANs 10, 20 and 40 existed on the access switches but not on the core switch, so their traffic was not being carried to the router.

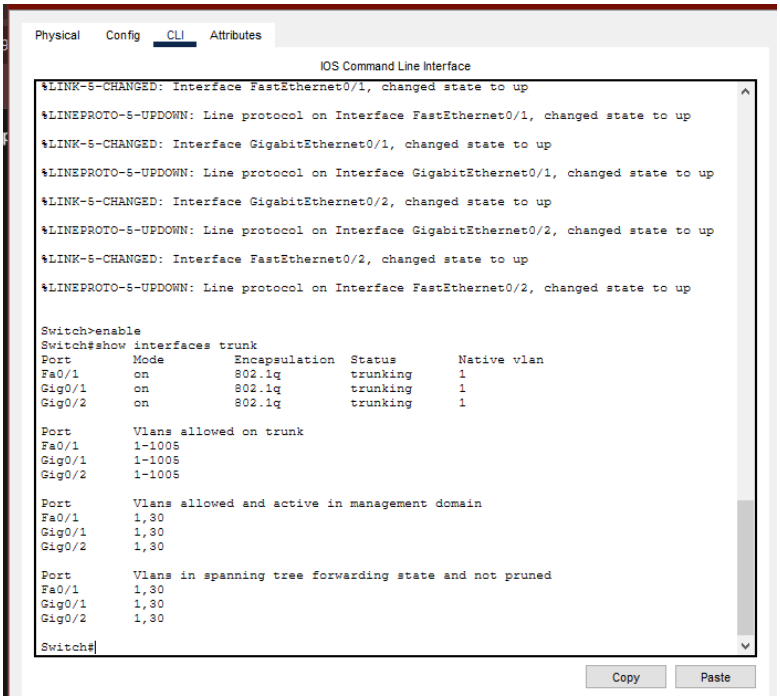


Figure 13: Before the fix: only VLANs 1 and 30 are active on the trunks.

Fix. VLANs 10, 20 and 40 were created on Switch0. The trunk output then listed VLANs 1, 10, 20, 30 and 40, and DHCP worked.

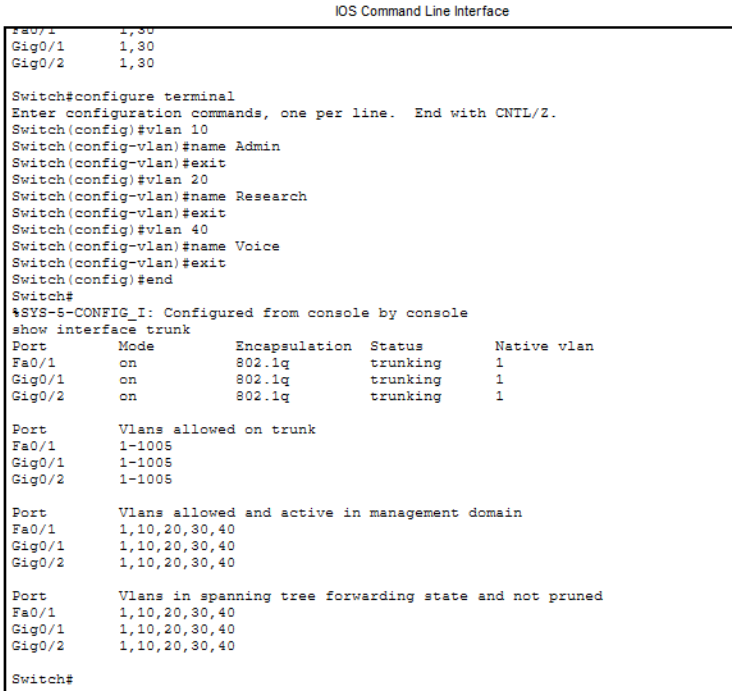


Figure 14: After the fix: all four VLANs are active on the trunks.

Problem 2: phone links stayed red. The 2960 switches do not provide Power over Ethernet, so the phones were not powered. Adding a power adapter module to each phone fixed this.

Problem 3: name resolution failed once. A DNS test failed because the name was typed incorrectly (kcbr instead of kbrc). Retyping it correctly resolved the name.

11. Reflection

This project showed me how the parts of a network depend on each other. The DHCP fault was the best example: the configuration on each device looked correct, but a VLAN missing from the core switch stopped traffic from reaching the router. I found it by checking the trunk status instead of guessing, which taught me to test layer by layer. I also learned to save versioned copies of my work after losing a Packet Tracer file to corruption and rebuilding the network. If I extended the design, I would add the redundancy described in Section 8.

12. Conclusion

The network meets the client's requirements: four VLANs with an addressing plan built from 192.168.48.0/24, inter-VLAN routing, automatic addressing by DHCP, a working internal DNS service (the assigned challenge), and VoIP phones accommodated under CR11. Testing confirmed each feature, and the single points of failure and recommended improvements are documented for the month-end downtime constraint.